2647203_P

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1 Lab Exercise

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[1]:
```
# Q1: Import the given "UScereal" dataset
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("UScereal.csv")
df.head()
```

[1]:

	Name	mfr	calories	protein	fat	sodium	fibre	carbo	sugars	shelf	potassium	vitamins
0	100% Bran	N	212.12	12.12	3.03	393.94	30.30	15.15	18.18	3	848.48	enriched
1	All-Bran	K	212.12	12.12	3.03	787.88	27.27	21.21	15.15	3	969.70	enriched
2	All-Bran with Extra Fiber	K	100.00	8.00	0.00	280.00	28.00	16.00	0.00	3	660.00	enriched
3	Apple Cinnamon Cheerios	G	146.67	2.67	2.67	240.00	2.00	14.00	13.33	1	93.33	enriched
4	Apple Jacks	K	110.00	2.00	0.00	125.00	1.00	11.00	14.00	2	30.00	enriched

[2]:
```
print(df.size)
print(df.shape)
```

```
780
(65, 12)
```

[3]:
```
# Q2: Find summary statistics of the dataset
df.describe()
```

[3]:

	calories	protein	fat	sodium	fibre	carbo
count	65.000000	65.000000	65.000000	65.000000	65.000000	65.000000
mean	149.408615	3.684000	1.422462	237.838308	3.870923	19.967538
std	62.411936	2.642821	1.647561	130.629537	6.133094	8.468190

```
min     50.000000   0.750000   0.000000    0.000000   0.000000  10.530000
25%    110.000000   2.000000   0.000000  180.000000   0.000000  15.000000
50%    134.330000   3.000000   1.000000  232.000000   2.000000  18.670000
75%    179.100000   4.480000   2.000000  290.000000   4.480000  22.390000
max    440.000000  12.120000   9.090000  787.880000  30.300000  68.000000

          sugars      shelf   potassium
count  65.000000  65.000000   65.000000
mean   10.051077   2.169231  159.119692
std     5.835252   0.839815  180.288575
min     0.000000   1.000000   15.000000
25%     4.000000   1.000000   45.000000
50%    12.000000   2.000000   96.590000
75%    14.000000   3.000000  220.000000
max    20.900000   3.000000  969.700000
```

[4]:
```
# Q3: Check for missing values
df.isnull().sum()
```

[4]:
```
Name         0
mfr          0
calories     0
protein      0
fat          0
sodium       0
fibre        0
carbo        0
sugars       0
shelf        0
potassium    0
vitamins     0
dtype: int64
```

[5]:
```
# Q4: Average protein value of each Manufacturer
df.groupby("mfr")["protein"].mean()
```

[5]:
```
mfr
G    2.885000
K    3.919048
N    7.026667
P    4.698889
Q    3.460000
R    2.604000
Name: protein, dtype: float64
```

[6]:
```
# Q5: Cereal with highest sugar from Manufacturer G
g_cereals = df[df["mfr"] == "G"]
```

```
high_sugar_cereal = g_cereals.loc[g_cereals["sugars"].idxmax()]
print("Cereal Name :", high_sugar_cereal["Name"])
print("Sugar Value :", high_sugar_cereal["sugars"], "g")
```

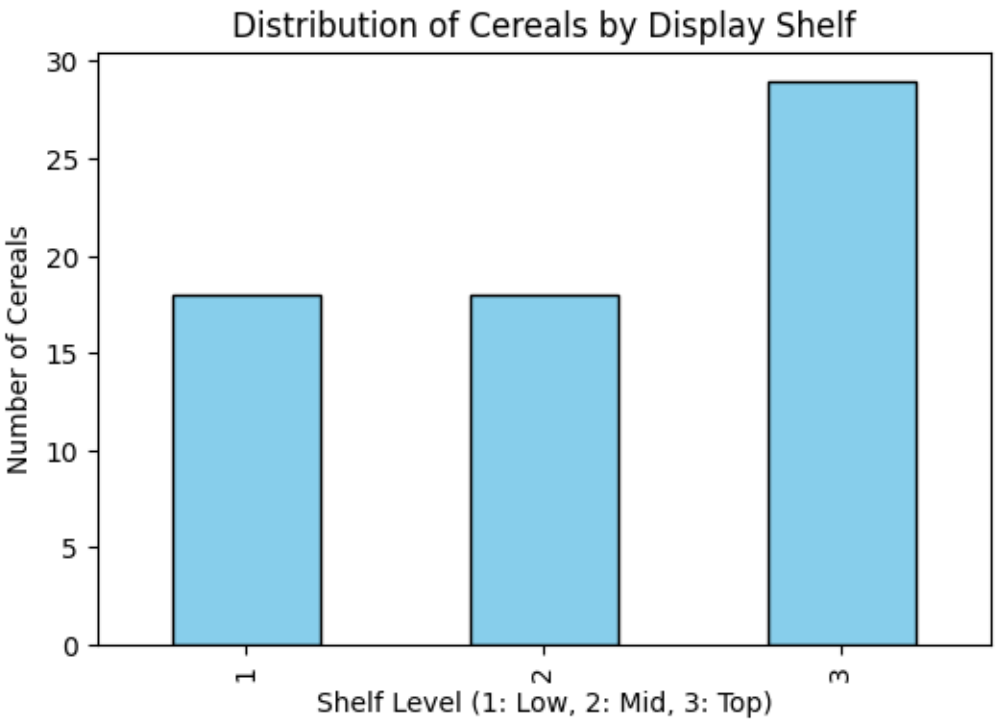
```
Cereal Name : Oatmeal Raisin Crisp
Sugar Value : 20.0 g
```

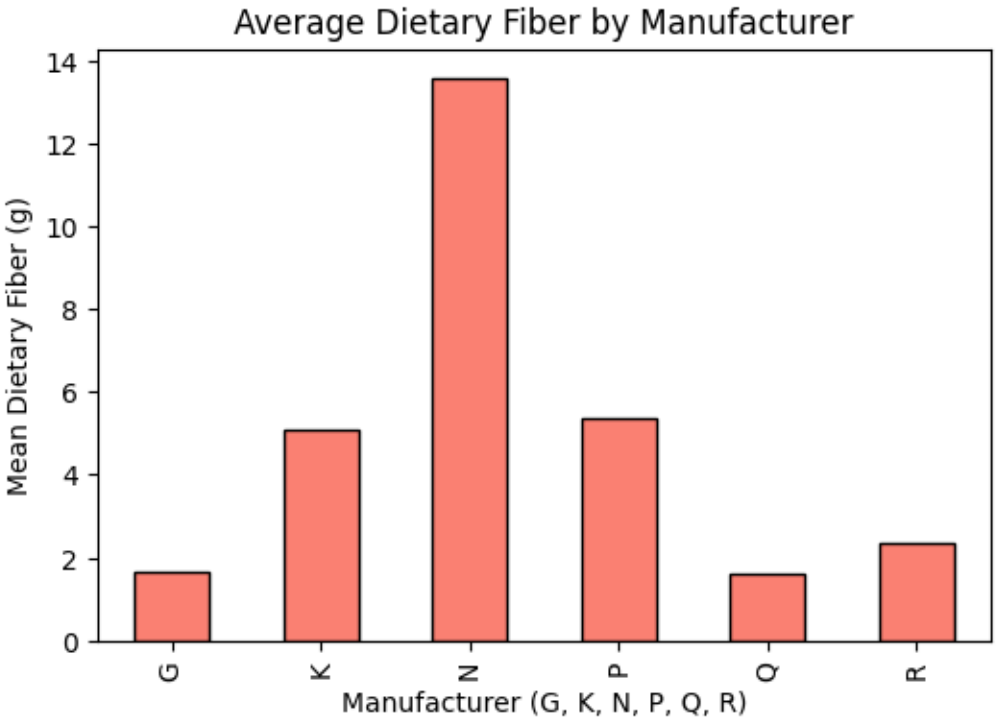
[7]:
```
# Q6: Data Visualizations

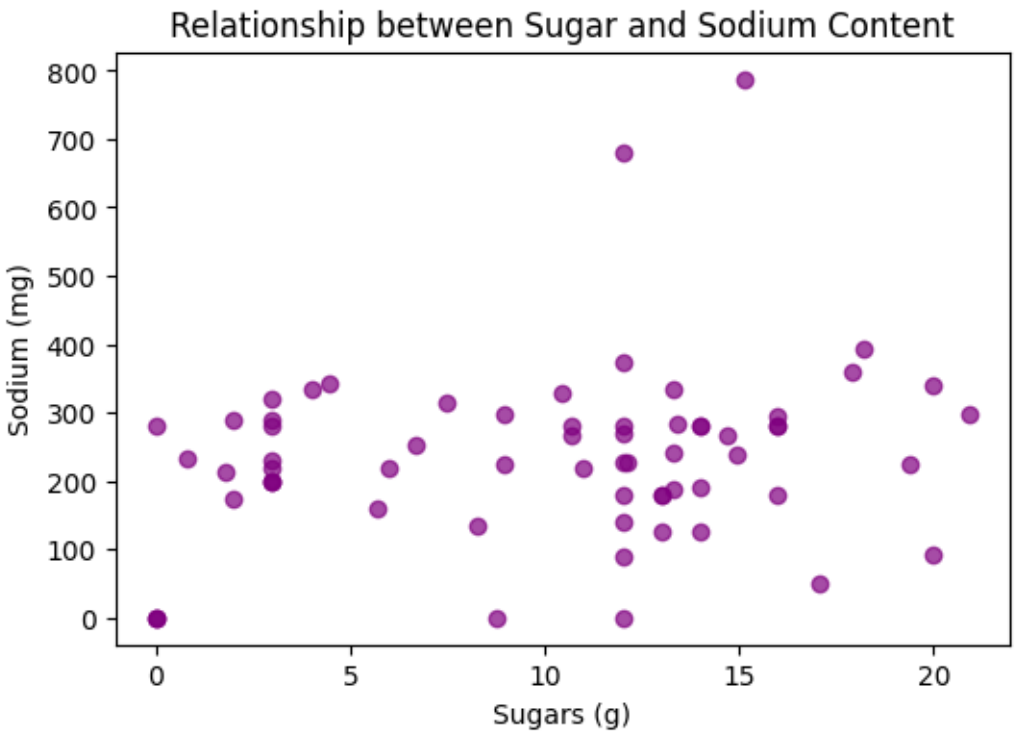
# (a) Count of cereals across display shelves (Bar plot)
plt.figure(figsize=(6, 4))
df["shelf"].value_counts().sort_index().plot(kind="bar", color="skyblue",␣
 ↪edgecolor="black")
plt.title("Distribution of Cereals by Display Shelf")
plt.xlabel("Shelf Level (1: Low, 2: Mid, 3: Top)")
plt.ylabel("Number of Cereals")
plt.show()

# (b) Average fiber content by manufacturer (Bar plot)
plt.figure(figsize=(6, 4))
df.groupby("mfr")["fibre"].mean().plot(kind="bar", color="salmon",␣
 ↪edgecolor="black")
plt.title("Average Dietary Fiber by Manufacturer")
plt.xlabel("Manufacturer (G, K, N, P, Q, R)")
plt.ylabel("Mean Dietary Fiber (g)")
plt.show()

# (c) Sodium vs Sugars (Scatter plot)
plt.figure(figsize=(6, 4))
plt.scatter(df["sugars"], df["sodium"], color="purple", alpha=0.7)
plt.title("Relationship between Sugar and Sodium Content")
plt.xlabel("Sugars (g)")
plt.ylabel("Sodium (mg)")
plt.show()
```


Distribution of Cereals by Display Shelf
Number of Cereals
0
5
10
15
20
25
30
1
2
3
Shelf Level (1: Low, 2: Mid, 3: Top)


Average Dietary Fiber by Manufacturer
Mean Dietary Fiber (g)
0
2
4
6
8
10
12
14
G
K
N
P
Q
R
Manufacturer (G, K, N, P, Q, R)


Relationship between Sugar and Sodium Content
Sodium (mg)
800
700
600
500
400
300
200
100
0
0
5
10
15
20
Sugars (g)

[ ]: